

Project ID :
25-26J-384
1. Topic (12 words max)

AI-Powered Drone System for Coconut Disease Detection, Real-time Monitoring and Intervention

2. Research group the project belongs to
SST - Software Systems & Technologies
3. Specialization of the project belongs to
Information Technology (IT)
4. If a continuation of a previous project:

Project ID	25-26J-384
Year	2025

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Sri Lanka's coconut industry is a vital contributor to the national economy and export sector [1]. However, effective monitoring of coconut tree health remains a significant challenge due to the industry's reliance on manual inspection methods. These traditional practices are labor-intensive, costly, and often fail to detect early signs of disease or nutrient deficiencies, leading to major crop losses and inefficiencies in chemical usage [2], [3].

Current visual inspection and basic remote sensing approaches used in plantations are highly prone to human error and offer limited spatial coverage [4]. The unique tropical environment of Sri Lanka, characterized by unpredictable monsoons, high humidity, and varying soil conditions, adds further complexity [6]. Diseases such as Coconut Cadang-Cadang and infestations like the Red Palm Weevil are widespread and require early detection, which is not adequately supported by existing systems [5], [7], [8].

Moreover, most coconut producers are small and medium-scale farmers who lack access to advanced monitoring technologies and the expertise to use them [9]. This gap between traditional farming and modern precision agriculture limits the industry's potential to improve productivity and sustainability [20], [21].

To address these challenges, we propose an Intelligent Multimodal Coconut Tree Health Monitoring System, designed specifically for Sri Lankan conditions. This system integrates drone-based surveillance with multispectral (NDVI, red-edge) and thermal imaging, alongside environmental data such as soil moisture and weather information, enabling comprehensive and accurate assessments even under adverse climatic conditions [10], [11], [12].

The system features:

Automated Tree Detection using YOLO-based object detection algorithms for identifying and tracking individual trees [13].

Disease Classification using CNN-ResNet models trained on local coconut disease datasets to identify symptoms early [7], [14].

Predictive Analytics that utilize historical disease data, growth trends, and environmental stress factors to forecast potential risks and recommend timely interventions [15], [16].

Edge Computing Architecture to allow real-time data processing on-site, reducing reliance on cloud services and ensuring faster response [17].

IoT Integration with sensors monitoring temperature, humidity, and soil conditions continuously [18].

User-Friendly Mobile Interface that employs NLP to enable farmers to interact through text, speech, or images, supporting a range of literacy and technology skills [19].

This intelligent monitoring system aims to revolutionize coconut cultivation in Sri Lanka by introducing precision agriculture methods adapted to local needs. It enhances early disease detection, optimizes input use, and provides actionable insights, empowering farmers with timely, data-driven decisions. By focusing on inclusivity and adaptability, it offers a scalable, sustainable solution for improving both productivity and resilience in the coconut sector.

References

- [1] Central Bank of Sri Lanka, "Annual Report 2023..."
- [2] K. Mahindapala et al., "Assessment of Coconut Cultivation Practices..."
- [3] W. Perera and S. Jayawardena, "Economic Impact of Coconut Disease Management..."
- [4] A. Rathnayake et al., "Remote Sensing Applications for Coconut Health Assessment..."
- [5] N. Dissanayake and R. Fernando, "Prevalence and Management of Coconut Cadang-Cadang Disease..."
- [6] T. Wickramasinghe et al., "Environmental Factors Affecting Coconut Health..."
- [7] S. Bandara et al., "Deep Learning Approaches for Coconut Disease Detection..."
- [8] M. Gunasekara and P. Silva, "Computer Vision Applications in Sri Lankan Coconut Disease Identification..."
- [9] L. Amarasinghe et al., "Multimodal Sensor Fusion for Tropical Crop Monitoring..."
- [10] D. Senanayake et al., "Precision Agriculture Implementation in Sri Lankan Coconut Estates..."
- [11] K. Ranasinghe and A. Weerasinghe, "Drone-Based Monitoring Systems..."
- [12] B. Jayasuriya et al., "YOLO-Based Coconut Tree Detection..."
- [13] H. Kumara et al., "ResNet Applications for Plant Disease Classification..."

- [14] R. Weerakoon and S. Pathirana, "Predictive Modeling for Coconut Yield Forecasting..."
- [15] G. Peiris et al., "Temporal Analysis of Coconut Disease Patterns..."
- [16] C. Rajapakse et al., "Environmental Stress Prediction Models..."
- [17] V. Mendis and K. Samaraweera, "Edge Computing Solutions for Real-time Crop Monitoring..."
- [18] C. Rajapakse et al., "IoT-Based Agricultural Monitoring..."
- [19] A. Herath et al., "Natural Language Processing Applications..."
- [20] Ministry of Agriculture, "National Agriculture Policy Framework 2020–2030..."
- [21] N. Abeysekera et al., "Sustainable Coconut Cultivation Practices..."

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

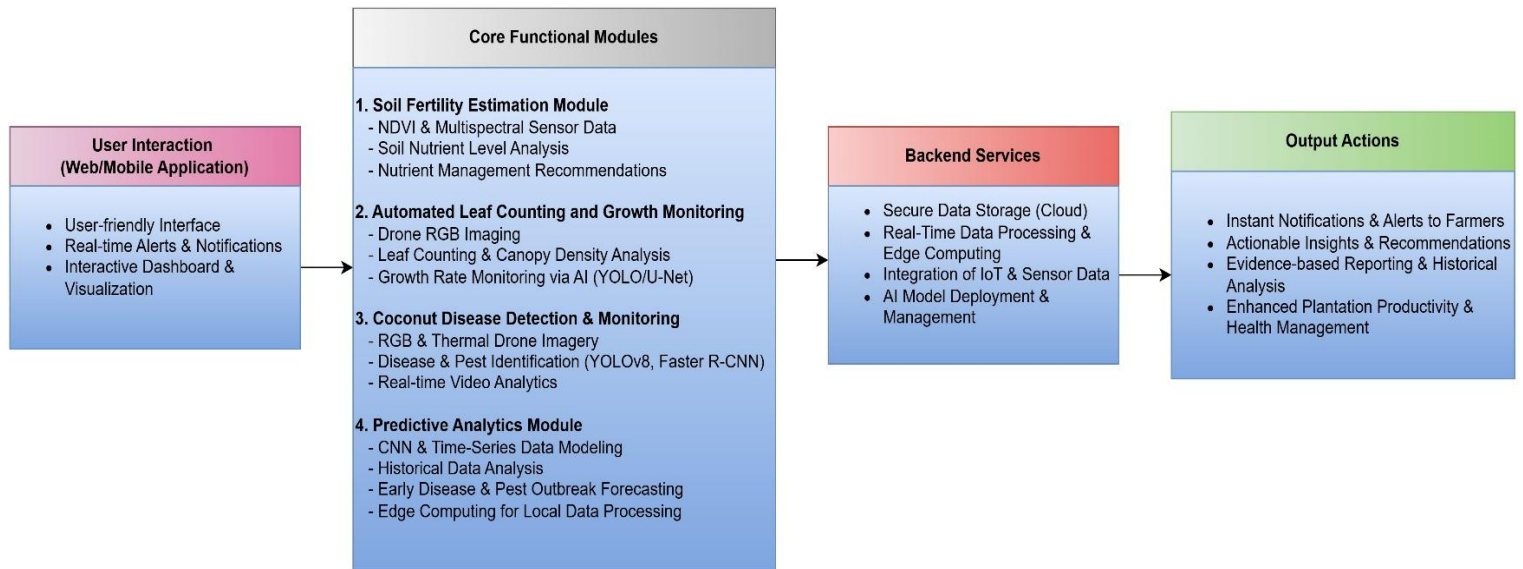
This innovative solution is designed as an intelligent and proactive agricultural management system, integrating artificial intelligence (AI), drone technology, sensor technology, and predictive analytics into a comprehensive and user-friendly platform tailored for coconut plantation management.

The solution utilizes drone-based RGB and thermal imaging to gather precise visual data from coconut plantations. Advanced computer vision and deep learning models, such as YOLOv8 and Faster R-CNN, accurately identify and classify diseases and pest infestations at early stages, enabling rapid intervention.

Multispectral sensors and NDVI analysis are employed to assess soil fertility and plant health conditions. Integrated IoT sensor technology continuously monitors soil moisture, nutrient levels, and environmental conditions, facilitating accurate and ongoing management.

The predictive analytics component employs CNNs and time-series forecasting models to anticipate risks of disease spread, nutrient deficiencies, and pest infestations. Edge computing processes this data locally, significantly enhancing response times and ensuring data privacy.

This solution provides farmers and plantation managers with an intuitive web and mobile interface featuring clear visualizations, instant notifications, and actionable insights. Ultimately, this system facilitates efficient decision-making, enhanced crop health, and increased productivity.



7. Brief description of specialized domain expertise, knowledge, and data requirements
(300 words max)

This project integrates specialized expertise across multiple domains within Information Technology, utilizing artificial intelligence (AI), machine learning (ML), computer vision, drone technology, sensor technology, and predictive analytics for efficient agricultural management. The primary objectives include soil fertility estimation, automated leaf counting and growth monitoring, coconut disease and pest detection, and predictive analytics for proactive interventions.

Expertise required includes proficiency in drone operations and capturing high-resolution RGB and thermal imagery. Skills in machine learning, particularly neural network training using frameworks such as TensorFlow, PyTorch, and OpenCV, are essential. Advanced knowledge in image processing and annotation methods for training deep learning models such as YOLOv8 and Faster R-CNN is critical.

Additional knowledge in multispectral imaging and NDVI analysis is required to accurately assess soil nutrient levels and plant health. Proficiency in IoT and sensor technology integration will enable real-time collection of soil moisture, nutrient levels, and environmental data.

For predictive analytics, specialized knowledge in convolutional neural networks (CNNs) for spatial pattern recognition and time-series analysis techniques is needed. Familiarity with edge computing to process data locally enhances privacy and responsiveness.

The project requires high-resolution RGB and thermal images, annotated datasets on coconut diseases, pest infestations, nutrient deficiencies, and growth stages. Environmental data collected through sensors, drone imagery, and historical records are fundamental for predictive modeling.

Expertise in UI/UX design, real-time data visualization, and system integration is essential to develop user-friendly mobile and web interfaces, providing farmers with accessible, efficient, and practical solutions.

8. Objectives and Novelty

Main Objective To design and develop an intelligent, integrated drone-based management system for coconut plantations, leveraging artificial intelligence, sensor technology, and predictive analytics to proactively monitor plant health, manage soil fertility, accurately detect diseases and pests, and provide early predictive alerts to enhance plantation productivity and sustainability.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Subasinghe S M U (IT22186638)	This system uses multispectral and thermal imaging to assess coconut tree health, detect stress and nutrient issues, create custom indices, and support precision farming with AI-powered mapping—boosting yield, cutting costs, and promoting sustainability	• Sensor configuration and calibration for coconut palms, flight planning with optimal coverage patterns, radiometric and geometric correction pipeline, spectral signature extraction and analysis. • NDVI calculation and threshold optimization, Enhanced Vegetation Index (EVI) implementation, Green NDVI and MCARI analysis, coconut-specific health indices creation.	• First comprehensive multispectral protocol for coconut palms, novel calibration methods for tropical environments, automated quality assessment for coconut canopy imaging • Development of Coconut Health Index (CHI), novel Frond Density Index (FDI), Nut Production Potential Index (NPPI), species-specific threshold calibration • Novel Water Stress Index for coconut palms, integration of thermal and multispectral data, early drought stress

		• Water Stress Index (WSI) calculation, thermal imaging integration, canopy temperature monitoring, drought tolerance assessment. • NPK deficiency mapping, micronutrient stress identification, deficiency symptom classification, nutrient uptake efficiency assessment. • Color-coded health map generation, productive vs. non-productive zone classification, stress severity scoring, interactive health dashboard creation. • Variable rate fertilizer mapping, targeted irrigation recommendations, treatment priority ranking, cost-optimization protocols.	detection system, irrigation scheduling optimization • Spectral signatures for coconut nutrient deficiencies, multi-nutrient deficiency detection algorithm, precision fertilizer recommendation system, boron and magnesium deficiency detection • Plantation health visualization • AI-powered stress classification for coconut palms • Automated health report generation • Predictive health trend analysisGPS-guided • precision application systems, cost-optimized treatment protocols, environmental impact minimization, ROI-based intervention planning
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Panditharathne R.L (IT22266200)	Implement a comprehensive predictive analytics framework leveraging convolutional neural networks (CNNs), time-series analysis, and edge computing technologies to accurately forecast risks associated with diseases, pests, and nutrient deficiencies in coconut plantations, and provide real-time actionable alerts.	• Collect historical data on coconut plantation health, including disease records, pest infestation histories, nutrient levels, and environmental factors such as temperature, rainfall, and humidity. • Perform detailed data pre-processing, including data cleaning, normalization, and aggregation to prepare high-quality datasets for analysis. • Develop CNN-based spatial analysis models to identify patterns of disease and pest occurrences from drone imagery. • Implement time-series forecasting models (LSTM, ARIMA) to predict temporal trends in diseases, pest infestations, and nutrient deficiencies. • Validate predictive model accuracy using historical records and sensor data.	• Combines CNN and time-series analysis to create a robust predictive analytics framework uniquely capable of accurately forecasting both spatial and temporal aspects of plantation health risks. • Integrates edge computing technologies for data processing, significantly enhancing response times while preserving user privacy. • Provides proactive and personalized early warnings, shifting from reactive disease and pest management to predictive and preventive interventions. • Leverages comprehensive environmental and historical datasets to deliver precise, actionable predictions and recommendations. • Enhances traditional agricultural management

		• Integrate edge computing solutions to process predictive analytics locally, ensuring faster response times and data privacy. • Create an intuitive, user-friendly web and mobile interface for farmers to receive instant predictive alerts and management recommendations.	through the seamless integration of advanced AI and predictive modeling, significantly reducing losses and improving overall plantation productivity and health.
Pasanjith W A R (IT22898920)	Develop an efficient drone and AI-driven system to detect, classify, and monitor the spread of diseases and pests in coconut plantations, providing farmers with actionable insights for timely interventions.	• Collect high-resolution RGB and thermal imagery of coconut plantations using drone technology. • Annotate and prepare datasets for common coconut diseases and pests, including Lethal Yellowing, Bud Rot, Red Palm Weevil, Rhinoceros Beetle, and Scale insects. • Implement image preprocessing techniques such as noise reduction, image enhancement, and GIS geo-referencing. • Train deep learning models like YOLOv8 and Faster R-CNN for accurate disease and pest identification.	• Integration of RGB and thermal imaging techniques for superior accuracy and early detection compared to traditional visual methods. • Advanced image enhancement methods optimized for aerial imagery captured under low-light and challenging environmental conditions. • Fully automated deep-learning software capable of detecting subtle disease and pest infestation patterns without human intervention. • Provision of a personalized mobile application featuring real-time

		• Perform video analysis with thermal imaging for early-stage disease detection. • Develop a web and mobile interface to provide farmers with immediate alerts and treatment recommendations.	alerts, intuitive navigation, clear visuals, and notifications, significantly enhancing farmer productivity and ease of use.
Karunaratne B.G.T.N (IT22027788)	To develop a soil fertility estimation system using multispectral data, and plant health indicators to assess nutrient levels and provide fertilizer recommendations for coconut plantations in Sri Lanka.	• Collect sensor-based data from coconut plantations. • Preprocess and normalize the collected data using image processing and data transformation techniques. • Develop an algorithm to correlate deficiencies with specific nutrient imbalances • Build a nutrient management recommendation engine that suggests corrective measures based on predicted soil fertility.	This module combines NDVI, multispectral imaging, and plant health data with machine learning to accurately estimate soil fertility and recommend nutrients. It offers a low-cost, automated solution tailored for small-scale coconut farmers in Sri Lanka, improving efficiency and reducing fertilizer waste.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Subasinghe S M U (IT22186638)	This spectral analysis system represents a comprehensive approach to monitoring and assessing coconut tree health using advanced multispectral imaging technology. The system integrates drone-based data acquisition with sophisticated analytical algorithms to evaluate various aspects of coconut palm health including vegetation vigor, water stress levels, nutrient deficiencies, and overall productivity. By capturing electromagnetic radiation across multiple wavelength bands, the system can detect subtle changes in tree physiology that are often invisible to the naked eye, enabling early intervention and precision management strategies. This technology transforms traditional plantation management by providing detailed, actionable insights that help optimize resource allocation, reduce operational costs, and maximize coconut yield while promoting sustainable agricultural practices.
Panditharathne R.L (IT22266200)	This component directly aligns with the Information Technology (IT) specialization by applying advanced analytical and technological principles central to IT, including machine learning, predictive analytics, time-series forecasting, convolutional neural networks (CNNs), and edge computing. It specifically leverages IT skills to develop intelligent solutions for proactive agricultural management within coconut plantations. The predictive analytics framework utilizes CNN-based models to effectively analyze spatial patterns from drone imagery data, enabling accurate identification of potential disease and pest infestations. Time-series models such as LSTM and ARIMA are employed to analyze historical plantation data, environmental parameters, and sensor information to accurately predict disease outbreaks, pest infestations, and nutrient deficiencies. Edge computing technology ensures local data processing, improving response times and data security by minimizing reliance on cloud-based servers. This integration demonstrates expertise in contemporary IT practices, particularly in managing data processing and ensuring robust privacy standards. Furthermore, the development of a user-friendly web and mobile application highlights proficiency in human-computer interaction, interface design, and data visualization, which are key competencies within the IT domain. Overall, this comprehensive use of advanced predictive analytics and cutting-edge IT infrastructure exemplifies the practical application of specialized IT knowledge to effectively address agricultural management challenges.

Pasanjith W A R (IT22898920)	This component directly aligns with the Information Technology (IT) specialization, incorporating fundamental IT principles such as artificial intelligence, machine learning, and computer vision. The system leverages advanced drone technology capable of capturing high-resolution RGB and thermal imagery to accurately detect diseases and pests affecting coconut plantations. Deep learning models such as YOLOv8 and Faster R-CNN are employed to swiftly identify and classify specific coconut diseases like Lethal Yellowing, Bud Rot, and pest infestations such as Red Palm Weevil and Rhinoceros Beetle. video analysis integrated with thermal imaging ensures proactive monitoring, significantly improving response times for necessary interventions. Furthermore, the development of user-friendly mobile and web interfaces, featuring intuitive navigation, clear visualization of disease conditions, and instant notifications, exemplifies the practical application of IT principles in user experience design and human-computer interaction. By integrating these advanced technologies effectively, this solution demonstrates a strong alignment with the practical and theoretical foundations of Information Technology, offering an efficient and scalable solution for agricultural management.
Karunaratne B.G.T.N (IT22027788)	This function uses data from NDVI and multispectral sensors, along with plant indicators like leaf color and growth rate, to evaluate the nutrient status of soil in coconut plantations. By analyzing this data with machine learning, the system identifies nutrient deficiencies (such as low nitrogen or potassium) and gives precise fertilizer recommendations for each area. This helps farmers improve tree health, reduce input costs, and increase yield efficiently.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Uthpala	Samarakoon	
Co-Supervisor	Ms.	Wishalya	Tissera	 25/06/2025
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	☒	No	☐
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- b) Does the proposed topic exhibit novelty?

Yes	☒	No	☐
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes	☒	No	☐
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	☒	No	☐
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

Approved

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.